

Signals & Systems (EE504)
Pre-Reading Material – Lecture 3
Elementary Signals in the Discrete Domain
Estimated Reading Time: 20–25 minutes

Course Context

This lecture introduces the fundamental building blocks of discrete-time signal analysis. It defines the **discrete-time unit step** and **unit impulse** signals and explores the behavior of **discrete-time exponential** signals. A significant portion of the lecture is devoted to understanding the **periodicity conditions** of discrete-time signals, which differ from those of continuous-time signals. Finally, the lecture introduces the basic system interconnection structures—**cascade**, **parallel**, and **feedback**—and classifies systems based on whether they possess **memory**.

Why This Topic Matters

Understanding discrete-time signals is the foundation of **Digital Signal Processing (DSP)** and **digital communication systems**. The sifting property of the impulse function provides the mathematical basis for **sampling**, the process of converting continuous-time signals into a form that digital systems can process. Furthermore, understanding how systems are interconnected (such as amplifiers connected in cascade) and how they store information or energy (memory) is essential for designing stable and efficient electronic systems.

Learning Outcomes

After completing this lecture, you should be able to:

- Define the discrete-time **unit step** ($u[n]$) and **unit impulse** ($\delta[n]$) signals.
- Explain the **sifting (sampling) property** of the continuous-time impulse function.
- Analyze the behavior of **discrete-time exponential** signals ($C\alpha^n$) for different values of α .
- Identify the valid range of **digital frequencies** and determine the **periodicity** of discrete-time sinusoidal signals.
- Distinguish between **cascade**, **parallel**, and **feedback** system interconnections.



- Classify systems as **memoryless (instantaneous)** or **dynamic (with memory)**.

Activate Your Prior Knowledge

In the previous lecture, you learned about continuous-time signal transformations and basic signals such as $u(t)$ and $\delta(t)$.

Recall that:

- $\delta(t)$ is an idealized impulse with infinite amplitude but unit area.
- $u(t)$ is a signal that switches from 0 to 1 at $t = 0$.
- Leonhard Euler's** relation,

$$e^{i\theta} = \cos\theta + j\sin\theta$$

provides the connection between complex exponentials and sinusoidal signals.

Prerequisite Concepts

- Rational Number:** A number that can be expressed as the fraction M/N , where M and N are integers. This concept is essential for determining the periodicity of discrete-time signals.
- Integer Variable (n):** In the discrete-time domain, the independent variable takes only integer values. Fractional indices such as $n = 0.5$ do not exist.
- Normalized Frequency (Ω_0):** The frequency variable used in the discrete-time domain, whose unique range lies between 0 and 2π .



Core Concepts

1. The Sifting Property

Multiplying an arbitrary signal $x(t)$ by an impulse $\delta(t-\tau)$ extracts (samples) the value of the signal at $t = \tau$. This property forms the mathematical foundation of digital sampling.

2. Discrete-Time Step and Impulse Signals

- $u[n]$ equals **1** for all $n \geq 0$ and **0** otherwise. It can be viewed as the sampled version of $u(t)$.
- $\delta[n]$ equals **1** only at $n = 0$ and **0** everywhere else. Unlike the continuous-time impulse, it has a finite amplitude of **1**.

3. Discrete-Time Exponential Signals (Ca^n)

- If $|\alpha| > 1$, the signal grows exponentially.
- If $|\alpha| < 1$, the signal decays exponentially.
- If α is negative, the signal alternates between positive and negative values.

4. Periodicity in Discrete Time

A discrete-time complex exponential,

$$e^{j\Omega_0 n}$$

is **periodic only if**

$$\Omega_0 = 2\pi\left(\frac{M}{N}\right),$$

where M and N are integers.

In other words, Ω_0 must be a **rational multiple of 2π** .

For example,

Prepared by: Dr. Rajesh Kumar

Course Repository: <https://github.com/rajesh101191/Signals-Systems-PIEMR-2026-EE504>

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$\cos(6n)$ is **not periodic** because its frequency is **not** a rational multiple of 2π .

5. System Interconnections

- **Cascade (Series/Tandem):** The output of one system becomes the input to the next system.
- **Parallel:** The same input is applied to multiple systems simultaneously, and their outputs are summed.
- **Feedback:** A portion of the output is fed back to the input, often to improve stability or performance.

6. Memory in Systems

- **Memoryless (Static):** The output depends only on the **present input** (e.g., a resistor).
- **Dynamic (With Memory):** The output depends on **past or future inputs**, or on stored energy (e.g., capacitors and inductors).

Guided Reading Questions

1. How is the relationship between $u[n]$ and $\delta[n]$ (difference) analogous to the relationship between $u(t)$ and $\delta(t)$ (derivative)?
2. Why does the frequency $\Omega_0 = 2\pi$ produce the same discrete-time signal as $\Omega_0 = 0$?
3. What mathematical condition must a discrete-time sinusoid satisfy to be periodic?
4. Can you name a common **hybrid system** used in modern electronics?
5. Why is a capacitor considered a **dynamic system**?

Reflection Activity

Think about the **audio volume control** on your computer.

1. Is it more likely to be a **memoryless system** or a **system with memory**?
2. If you add an **echo effect**, does your answer change? Why?



3. How would you represent a **two-stage amplifier** using the system interconnection types discussed in this lecture?

Self-Assessment

True or False

1. The amplitude of a discrete-time unit impulse $\delta[n]$ is infinitely large.

Answer: False (*Its amplitude is exactly 1.*)

2. Every discrete-time cosine signal is periodic.

Answer: False (*A discrete-time cosine is periodic only if its frequency is a rational multiple of 2π .*)

Short Answer

What is the difference between a cascade and a parallel system connection?

Answer: In a **cascade** connection, systems are connected in series, so the output of one system becomes the input of the next. In a **parallel** connection, the same input is applied to multiple systems simultaneously, and their outputs are added together.

Think About It

If the unique frequency range of a discrete-time signal extends only from **0 to π** , what happens to the **rate of oscillation** as the frequency increases from **π** toward **2π** ?

